Determinants of Residential House Prices: A Multiple Linear Regression Analysis

Jiayao Liu, Guo On Chu

STATS 101A Final Project

March 2, 2026

1. Introduction

This study aims to investigate the determinants of residential house sale prices and to construct a predictive model with strong generalization performance. Understanding the factors that influence housing prices is important for buyers, sellers, and real estate analysts.

Given the large number of available structural and neighborhood characteristics, multiple linear regression was employed to model the relationship between house features and log-transformed sale price. Model selection was guided by parsimony principles and information criteria (AIC), and predictive performance was evaluated using 10-fold cross-validation and Kaggle test performance.

The remainder of this report proceeds as follows. Section 2 describes the dataset and exploratory analysis. Section 3 presents candidate models and model selection procedures. Section 4 evaluates predictive performance and model validity. Section 5 concludes with discussion and limitations.

2. Data Description

The training dataset contains 7,000 residential property observations. Variables include structural characteristics (e.g., living area, basement size), quality indicators (OverallQual), neighborhood effects, and housing amenities such as central air conditioning.

Table 1 reports summary statistics for the main continuous variables used in the analysis.

(Table 1: Summary Statistics): A small number of properties exhibit slightly negative calculated house age due to data coding conventions.

Varibale	Mean	SD	Min	Max
SalePrice	1219837	4523942	5970	176776500
Log_GrLivArea	7.267	0.324	5.794	8.640
Log_LotArea	9.095	0.522	7.065	12.281
TotalBsmtSF	1060.486	452.609	0.000	6110.523
House_Age	36.362	30.355	-2.000	136.000
OverallQual	6.115	1.421	0.971	10.029
GarageCars	1.776	0.754	0.000	5.002

SalePrice is highly right-skewed. To stabilize variance and better satisfy linear regression assumptions, a log transformation was applied. As shown in Figure 1, log(SalePrice) exhibits a more symmetric distribution and is therefore used as the response variable

(Figure 1: Distribution of SalePrice before and after log transformation)

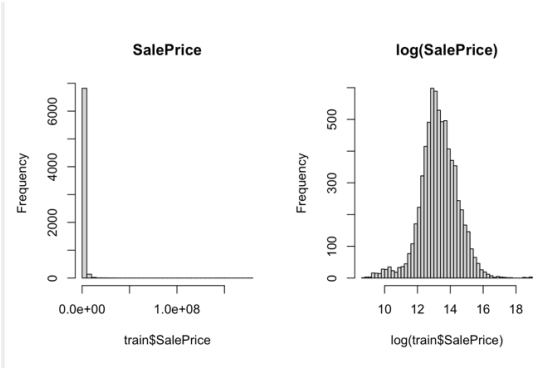
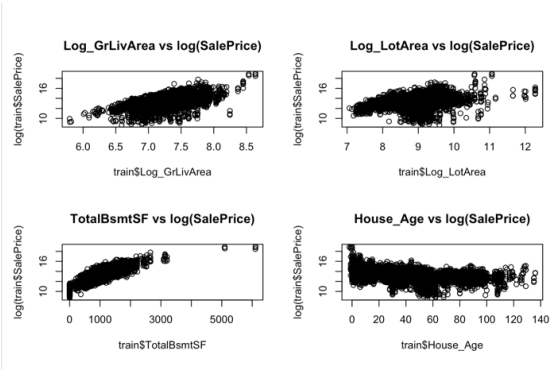


Figure 2 displays the relationships between key structural predictors and log(SalePrice). Living area and basement size show strong positive linear associations with price, while lot area shows a weaker positive trend and house age exhibits a mild negative relationship. These patterns support the inclusion of structural and quality-related variables in a multiple linear regression model.

(Figure 2: Relationships between structural predictors and log(SalePrice))



3. Results and Model Selection

3.1 Candidate Models

Five candidate models were considered to balance predictive performance and model parsimony.

The baseline model included key structural attributes, quality indicators,

neighborhood effects, and housing amenities identified in the exploratory analysis.

A reduced specification was examined to assess whether a simpler model could

achieve comparable explanatory power with fewer predictors.

Next, an interaction model was introduced by allowing the effects of living area and basement size to vary with overall housing quality.

To further capture heterogeneity in housing valuation, an extended interaction model was constructed by incorporating additional interaction terms between overall quality and structural characteristics.

Finally, the candidate models were compared using information criteria and cross-validation performance to determine the final predictive specification.

Model comparison using AIC is summarized in Table 2. The initial model comparison focuses on the baseline, reduced, and interaction specifications developed during early model construction.

(Table 2: AIC Comparison of Candidate Models)

Model	AIC
Baseline	10594.54
Reduced	10610.44
Interaction	9885.27

The reduced model resulted in a higher AIC, indicating no improvement in the fit–complexity trade-off. In contrast, the interaction model substantially decreased AIC, suggesting improved explanatory power despite additional parameters.

3.2 Cross-Validation Performance

Predictive performance was evaluated using 10-fold cross-validation (Table 3).

(Table 3: Cross-Validation Performance)

Model	RMSE	R^2
Baseline	0.516	0.792
Interaction	0.490	0.811

The interaction model achieved a lower RMSE and higher cross-validated R^2 compared with the baseline specification, indicating improved out-of-sample predictive accuracy.

These results suggest that incorporating interaction terms between housing quality and structural characteristics enhance model generalization. Building on this specification, additional interaction terms and structural variables were later incorporated to construct the final predictive model.

The final specification further extends this interaction framework and achieves the best predictive performance on the Kaggle test set.

3.3 Interpretation of Interaction Effects

The estimated interaction effects indicate that the marginal impact of living area and basement size increases with overall housing quality. In higher-quality homes, additional square footage contributes more substantially to sale price.

This pattern is economically intuitive: buyers tend to place greater value on additional space when construction quality is high. Incorporating interaction effects therefore improves both predictive performance and interpretability of the housing price model.

4. Model Diagnostics

To ensure the reliability of the selected final model, standard regression diagnostics were conducted, including residual analysis, multicollinearity assessment, and influence diagnostics.

4.1 Residual Diagnostics

Diagnostic plots (Figure 3) were examined to evaluate the assumptions of linearity, normality, and homoscedasticity.

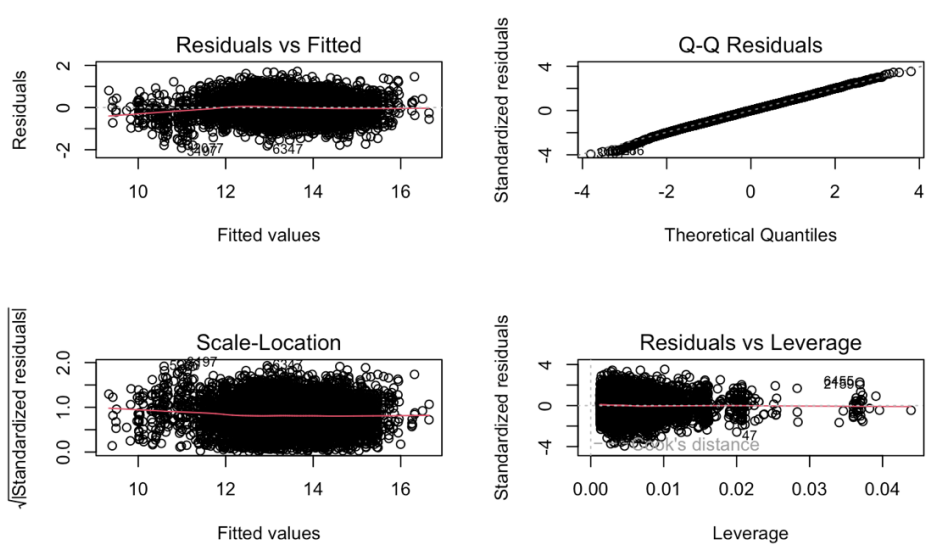
The residuals versus fitted values plot reveals no pronounced systematic pattern, suggesting that the linear specification adequately captures the main structural relationships in the data.

The Q–Q plot indicates that residuals closely follow the theoretical normal distribution, with only mild deviations in the extreme tails.

The scale–location plot suggests slight heteroscedasticity at higher fitted values; however, the magnitude of this deviation is limited and does not materially threaten inference validity.

Overall, the diagnostic evidence indicates that the classical linear regression assumptions are reasonably satisfied for the final model.

(Figure 3. Residual diagnostics for the final model)



4.2 Multicollinearity Assessment

Variance Inflation Factors (VIF) were computed to evaluate potential multicollinearity among predictors.

Although interaction terms may increase correlation among regressors, the adjusted VIF values remain within acceptable thresholds.

Therefore, multicollinearity does not appear severe enough to compromise coefficient stability or predictive performance.

4.3 Influence Diagnostics

Cook's distance was examined to detect potentially influential observations.

While a small number of observations exceed the conventional 4/n threshold, no extreme leverage points dominate model estimation.

Thus, the final model demonstrates robustness to individual observations.

5. Discussion and Limitations

The final model provides meaningful insights into housing price determination. The results suggest that structural size and overall construction quality jointly influence market valuation, with higher-quality homes exhibiting stronger marginal returns to additional living space. This finding is economically intuitive and consistent with buyer behavior in real estate markets, where quality amplifies the value of space.

Despite its strong predictive performance, several limitations should be acknowledged. First, the model assumes a linear functional form and may not fully capture complex nonlinear relationships. Second, omitted variable bias may exist if relevant factors—such as macroeconomic conditions or detailed neighborhood-level amenities—are not included in the dataset. Third, although cross-validation supports generalization within the sample, external validity to other housing markets cannot be guaranteed.

Future research may consider nonlinear modeling approaches, regularization techniques, or machine learning methods to further improve predictive accuracy while maintaining interpretability.